

Explainable Machine Learning for District-Level Rural Employment Vulnerability and Unmet MGNREGA Demand Forecasting in India

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Abstract—The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) legally entitles over 83.8 million rural households to 100 days of unskilled wage employment annually. However, administrative rationing, fiscal liquidity delays, and localized agro-meteorological shocks leave over 14.02 million households with unmet employment demand, yielding a national average fulfillment rate of 83.3%. This paper develops an empirical Explainable Machine Learning (XAI) and Decision-Support Framework across 745 Indian districts in 34 states and union territories. We formulate a multimodal 5-pillar data integration architecture unifying: (1) official Ministry of Rural Development administrative MIS records; (2) NFHS-5 female schooling and literacy metrics; (3) NITI Aayog Multidimensional Poverty Index (MPI) deprivation scores; (4) IMD gridded daily precipitation aggregates; and (5) Ministry of Panchayati Raj Local Government Directory (LGD) spatial crosswalk keys. To address asymmetric welfare risks, we formulate Quantile Gradient Boosting Regressors (P_{05}, P_{50}, P_{95}) for 90% confidence bands alongside an ensemble Random Forest Classifier. On 5-fold cross-validation, Gradient Boosting achieves an R^2 of 0.7001 ± 0.059 (RMSE = 8,896 households), outperforming Random Forest (0.6964), Ridge Regression (0.6096), and Decision Trees (0.5396). The Random Forest Classifier delivers 79.19% test accuracy ($F_1 = 0.7861$). Local feature force attributions isolate Work Demand Pressure (48.2%), SC/ST Marginalization (14.1%), and compound Literacy $\times$ Climate Stress (12.8%) as primary distress drivers. An empirical out-of-domain transferability experiment between Peninsular South-West and Gangetic Heartland India reveals sharp R^2 degradation (to 0.552 and 0.518), demonstrating that national welfare models require decentralized regional calibration. Finally, by mapping forecasts to FY 2024–25 notified wage rates, we formulate a 45-day district-level fiscal buffer model identifying an annual national contingency requirement of ₹17,324.5 Crores. The complete decision-support engine is open-sourced and deployed live on Netlify.

Index Terms—MGNREGA, Rural Employment Vulnerability, Explainable Artificial Intelligence (XAI), Quantile Regression, Local Feature Attributions, Climate Shocks, LGD Harmonization, Public Policy Analytics.

I. INTRODUCTION

Public employment guarantees serve as vital macroeconomic stabilizers across agrarian developing economies. Enacted in 2005, the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) legally guarantees 100 days of unskilled wage work per year to any rural household volunteering to perform manual work. Covering over 143 million active registered workers across 700+ districts with an annual budget exceeding ₹86,000 Crores (\$10.5B USD), MGNREGA represents the largest statutory safety-net program globally.

The Rationing Conundrum: Although legally conceived as an open-ended rights guarantee, the program routinely transforms into a rationed supply-constrained regime. Bureaucratic fund delays, administrative bottlenecks, and climate shocks leave over **14.02 million households without work** annually. Crucially, conventional research relies on Decennial Census 2011 statistics, ignoring dynamic human capital advances and recent district bifurcations. Furthermore, standard mean-squared error (OLS) algorithms produce point forecasts that fail to quantify the asymmetric risks of public distress. This paper develops an empirical, multi-pillar **Explainable Machine Learning (XAI) Decision-Support Engine** across 745 Indian districts that provides 90% predictive intervals, transparent causal attributions, regional generalizability audits, and fiscal contingency buffers.

II. RELATED WORK & EMPIRICAL GAPS

The economic literature has rigorously analyzed MGNREGA's consumption-smoothing effects (Subbarao 2003; Drèze & Khera 2017) and administrative leakages (Sukhtankar 2016; Muralidharan et al. 2016). In parallel, climate economists have highlighted the impact of erratic monsoonal precipitation on rural farm distress (Taraz 2017). However, existing studies lack unified crosswalk harmonization, treat climate shocks linearly, and fail to provide explainable machine learning decision support. Table I systematically contrasts the state of the art.

TABLE I: RESEARCH GAP MATRIX & METHODOLOGICAL INNOVATIONS

Gap ID	Research Literature Gap	Conventional Approach	Proposed Methodological Solution
GAP 1	Target Resolution	State aggregate budgets	745 districts household unmet demand
GAP 2	Boundary Bifurcations	Drop bifurcated units / Census '11	Ministry LGD Master Code crosswalk
GAP 3	Human Capital Dynamism	Static Census 2011 literacy	NFHS-5 female schooling & literacy metrics
GAP 4	Deprivation Dimensions	Unidimensional consumption	NITI Aayog MPI 12-indicator deprivation
GAP 5	Climate Shock Granularity	State annual rainfall sums	IMD gridded monthly & monsoon precipitation
GAP 6	Compound Stress Modeling	Linear additive features	Non-linear Literacy $\times$ Climate Stress Index
GAP 7	Welfare Risk Asymmetry	Symmetric OLS point estimates	Quantile GBR (P05, P50, P95) 90% intervals
GAP 8	Local Interpretability	Opaque black-box models	SHAP-style local feature forces per district
GAP 9	Operational Fiscal Linking	Theoretical labor models	State wage-notified 45-day contingency buffer
GAP 10	Regional Transferability	Assumed national invariance	Empirical out-of-domain transfer experiment

III. 5-PILLAR DATA ARCHITECTURE & SPATIAL HARMONIZATION

Our data architecture fuses five official sources: (1) MoRD MGNREGA MIS administrative logs (740 districts); (2) NFHS-5 factsheet indicators on female literacy and schooling; (3) NITI Aayog Multidimensional Poverty Index (MPI) scores; (4) IMD Pune 0.25° daily gridded precipitation summaries; and (5) Ministry of Panchayati Raj Local Government Directory (LGD) codes. Standardized uppercase regular expressions resolve spelling discrepancies across portals, followed by LGD code mapping.

Mathematical Formulations of Engineered Features:

1) *Demand Pressure Ratio (DPR)*: Measures the proportion of registered rural households actively seeking statutory wage work:

$$DPR_i = HH_Demanded_i / (Jobcards_Total_i + 1) \quad (1)$$

2) *SC/ST Marginalization Share (MS)*: Quantifies the proportion of historically disadvantaged caste groups:

$$MS_i = (Jobcards_SC_i + Jobcards_ST_i) / (Jobcards_Total_i + 1) \quad (2)$$

3) *Literacy $\times$ Climate Stress Index (LCSI)*: Non-linear compound vulnerability formulation:

$$LCSI_i = [(100 - Female_Literacy_i) \times 100] / (Rainfall_{annual,i} + 50) \quad (3)$$

4) *Human Capital Deprivation Index (HCDI)*: Weighted living standards deprivation:

$$HCDI_i = 0.40(100 - Literacy) + 0.30(100 - Sanitation) + 0.30(100 - Clean_Fuel) \quad (4)$$

IV. PREDICTIVE MACHINE LEARNING METHODOLOGY

A. Quantile Loss Formulation: Standard regression minimizes symmetric squared error. In social safety nets, under-predicting distress leads to starvation and despair, whereas over-allocation incurs minimal societal cost. We formulate Quantile Gradient Boosting using pinball loss:

$$\rho_q(y, \hat{y}) = \max(q(y - \hat{y}), (1 - q)(\hat{y} - y)), \quad q \in (0, 1) \quad (5)$$

We train three models ($q \in \{0.05, 0.50, 0.95\}$) establishing the 90% confidence interval $[\hat{I}_{p05}, \hat{I}_{p95}]$.

B. Multi-Tier Risk Classification: Districts are classified into High (Fulfillment < 80% or Unmet > 25k), Medium (80% – 92%), and Low ($\geq 92\%$) risk using an ensemble Random Forest with Gini split criteria.

C. Local Feature Force Attributions: District-level SHAP proxy forces are computed via:

$$\phi_{i,k} = 10 \times [(x_{i,k} - \text{Median}(X_k)) / \sigma(X_k)] \times \omega_k \quad (6)$$

where ω_k represents the global tree split importance.

D. Contingency Wage Buffers: Budgetary reserves are computed as:

$$C_i = [\text{P}_{50}(x_i) \times 45 \times \text{Wage}_{\text{state}(i)}] / 10^7 \quad (\text{in } \text{Crores}) \quad (7)$$

V. EXPERIMENTAL BENCHMARKS & TRANSFERABILITY

All models were validated using 5-fold cross-validation on 745 districts.

TABLE II: REGRESSION BENCHMARKS (PREDICTING UNMET DEMAND HH)

Model Architecture	Test R^2	5-Fold CV R^2 ($\mu \pm \sigma$)	RMSE (HH)	MAE (HH)
Gradient Boosting Regressor (GBR)	0.6988	0.7001 $\pm$ 0.0596	8,896.8	5,519.2
Random Forest Regressor (RFR)	0.6940	0.6964 $\pm$ 0.0359	8,967.5	5,484.2
Ridge Regression (L2 Regularized)	0.6394	0.6096 $\pm$ 0.0482	9,735.6	7,000.3
Decision Tree Regressor	0.5948	0.5396 $\pm$ 0.1538	10,320.1	6,540.5

TABLE III: CLASSIFICATION BENCHMARKS (VULNERABILITY TIERS)

Model Architecture	Accuracy (%)	5-Fold CV Acc ($\mu \pm \sigma$)	Precision (Wtd)	F1-Score (Wtd)
Random Forest Classifier	79.19%	72.89% $\pm$ 2.80%	0.7912	0.7861
Support Vector Classifier (SVC)	77.18%	71.28% $\pm$ 3.10%	0.7710	0.7687
Gradient Boosting Classifier	76.51%	73.29% $\pm$ 2.90%	0.7645	0.7637
Multinomial Logistic Regression	69.13%	68.46% $\pm$ 3.40%	0.6850	0.6803

TABLE IV: CROSS-REGIONAL MODEL TRANSFERABILITY EXPERIMENT (GAP 15)

Training Macro-Region	Target Testing Macro-Region	In-Domain R^2	Out-of-Domain R^2	Out-of-Domain Acc
South & West (Peninsular)	North & Central (Heartland)	0.694	0.552	59.3%
North & Central (Heartland)	South & West (Peninsular)	0.702	0.518	61.4%

Transferability Finding: Out-of-domain performance degrades significantly (R^2 drops from ~ 0.70 to 0.55 and 0.52). South/West models underestimate high-volume seasonal agricultural distress in the Gangetic plain, whereas North/Central models over-predict unmet demand in peninsular states with diversified non-farm rural economies. National uniform welfare algorithms are inadequate; regional calibration is mandatory.

VI. EXPLAINABILITY & FISCAL POLICY IMPLICATIONS

Global Feature Importance: Work Demand Pressure accounts for **48.2%** of predictive importance, followed by SC/ST Marginalization Share (**14.1%**), Literacy $\times$ Climate Stress (**12.8%**), Female Literacy Rate (**8.4%**), Annual Rainfall (**7.2%**), Basic Deprivation (**5.3%**), and Registered Jobcards (**4.0%**).

Case Analysis: In Bundelkhand (e.g., Mahoba, Banda), positive forces are dominated by climate deficit ($\phi^{\text{Climate}} = +4.8$) and literacy gaps ($\phi^{\text{Literacy}} = +3.2$), pushing P95 crisis demand to 49,200 HHs. Conversely, in the Eastern Gangetic Basin (e.g., Sitamarhi, Bahaich), distress is driven by massive administrative demand pressure ($\phi^{\text{Demand}} = +8.4$) outstripping muster capacity.

Fiscal Safety-Net Budgeting: Across 745 districts, the 45-day contingency fund model identifies an aggregate national buffer requirement of **17,324.5 Crores**, providing finance departments with an empirical benchmark for emergency wage releases.

VII. INTERACTIVE DECISION SUPPORT SYSTEM

The complete decision support architecture is deployed live on Netlify at: <https://explainablemlfordistrictlevelemp.netlify.app/>. The client-side dashboard integrates Leaflet geospatial mapping, Chart.js metric visualizations, interactive 90% confidence interval explorers, what-if policy simulators, and an embedded AI Policy Copilot. All raw datasets, ingestion scripts, models, and web assets are fully documented and version-controlled at <https://github.com/prachigarg1511/ExplainableMLForDistrictLevelRuralEmpMGNREGA>.

VIII. CONCLUSION & POLICY RECOMMENDATIONS

This research establishes an explainable, multimodal machine learning framework for district-level rural employment vulnerability and unmet MGNREGA demand forecasting in India. By integrating five administrative, climatic, and human capital pillars across 745 districts, formulating quantile prediction intervals, providing localized causal force decompositions, and stress-testing cross-regional transferability, this work bridges academic machine learning and proactive public welfare administration.

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